Surrogate Modeling of CFD Flow Fields using Graph Attention Networks on the AirfRANS Dataset

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Introduction

Computational Fluid Dynamics (CFD) is widely used to analyze aerodynamic flow fields, but high-fidelity RANS simulations require significant computational cost. To reduce computation time, machine learning surrogate models have been proposed to approximate CFD results efficiently. In this study, we build upon the ML4CFD competition framework using the AirfRANS dataset. We apply a Graph Attention Network (GAT) to predict flow field variables from mesh-based graph data. The objective of this work is to evaluate the effectiveness of graph neural networks for surrogate modeling of CFD simulations.

AirfRANS Dataset

AirfRANS is a public benchmark dataset providing CFD simulation results of 2D RANS steady-state, incompressible airflow around airfoil geometries.

Mesh nodes from OpenFOAM: simulations serve as the point cloud input to the model.

Dataset Splits

- **Training**: 200 simulations
- **Testing**: 200 simulations (in-distribution)
- **OOD Test**: 496 simulations (unseen Re)

Inputs (per mesh node)

- Position (x, y)
- Inlet Velocity (u, v)
- Surface Normal (n_x, n_y)
- Distance to Surface (d)

Outputs (target flow fields)

- Velocity (u, v)
- Pressure/Density (P/ρ)
- Turbulent Viscosity (ν_t)

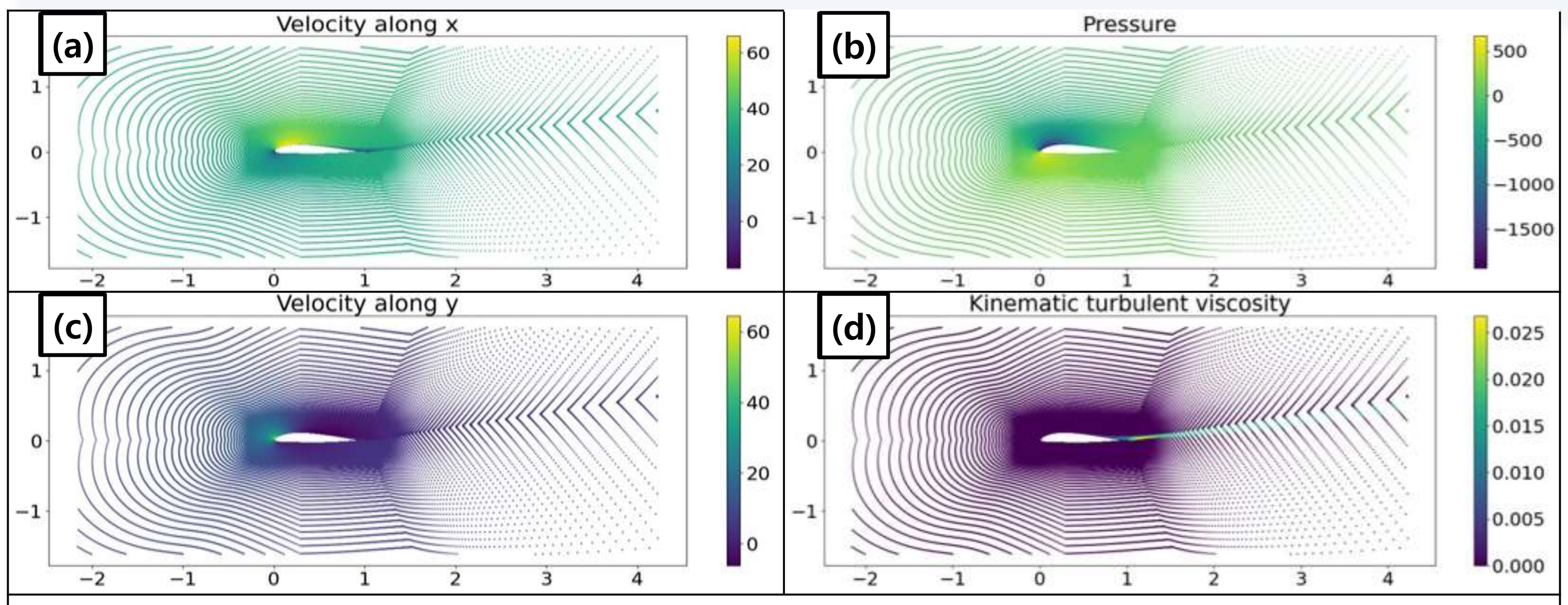

Fig. 1 Sample Simulation output (a) Velocity — x component [m/s] (b) Pressure [Pa/ρ] (c) Velocity — y component [m/s] (d) Kinematic turbulent viscosity [m²/s]

Reynolds- Averaged Navier-Stokes equations

The Reynolds-averaged Navier–Stokes (RANS) approach decomposes the instantaneous velocity into a mean and fluctuating component: $U_i = \overline{U}_i + u_i$

Averaging the Navier–Stokes equations yields the Reynolds equations:

$$\frac{\partial \overline{U}_j}{\partial t} + \overline{U}_j \frac{\partial \overline{U}_i}{\partial x_j} = \nu \nabla^2 \overline{U}_j - \frac{\partial \overline{u_i u_j}}{\partial x_j} - \frac{1}{\rho}\frac{\partial \overline{p}}{\partial x_j}$$

The Reynolds stress tensor $-\overline{u_i u_j}$ introduces the closure problem: it cannot be derived from mean quantities alone. Under the Boussinesq turbulent-viscosity hypothesis:

$$-\overline{u_i u_j} + \frac{2}{3} k \delta_{ij} = \nu_T \left(\frac{\partial \overline{U}_i}{\partial x_j} + \frac{\partial \overline{U}_j}{\partial x_i} \right)$$

The turbulent viscosity ν_T is computed via the k–ω SST model, making it one of the four flow fields our surrogate model must predict.

Method

k-NN Graph Construction

CFD meshes are unstructured point clouds. **k-Nearest Neighbor** graphs are built per simulation based on Euclidean distance in (x, y) space.

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Edge (i, j)  ⟺  j ∈ kNN(i)
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- Undirected — both directions kept
- Rebuilt independently per simulation
- Distance metric: $\|pos_i - pos_j\|_2$

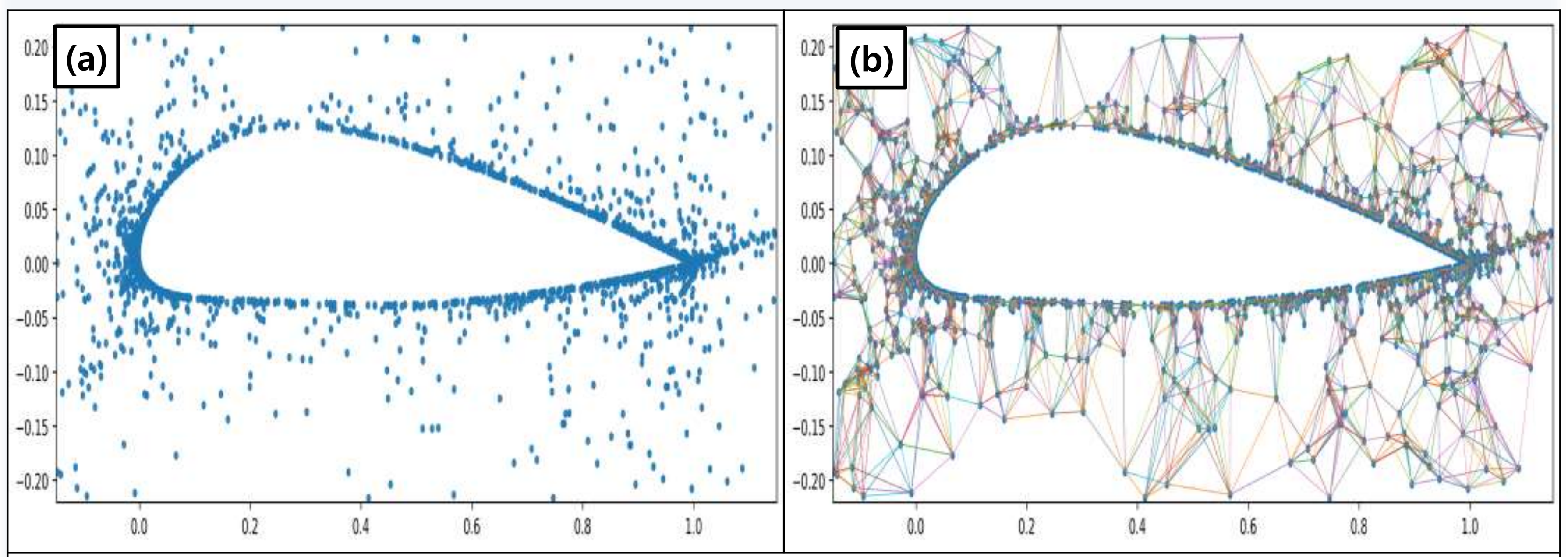

Fig. 2 (a) Unstructured point cloud representation of the CFD mesh. (b) k-NN graph (k=8) capturing local connectivity.

Graph Attention Network (GAT)

GAT assigns learnable attention weights to each neighbor node, focusing on the most relevant context:

Step 1 — Attention score

$$e_{ij} = a \cdot [\, W h_i \,\|\, W h_j \,]$$

W: weight matrix a: attention vector ‖: concat

Step 2 — Normalize (masked softmax)

$$\alpha_{ij} = \text{softmax\_j}(\,\text{LeakyReLU}(e_{ij})\,)$$

Only j ∈ N(i) included masked attention

Step 3 — Multi-head Aggregation

$$\vec{h}_i' = \|_{k=1}^{K} \sigma \left(\sum_{j \in \mathcal{N}_i} \alpha_{ij}^k W^k \vec{h}_j \right)$$

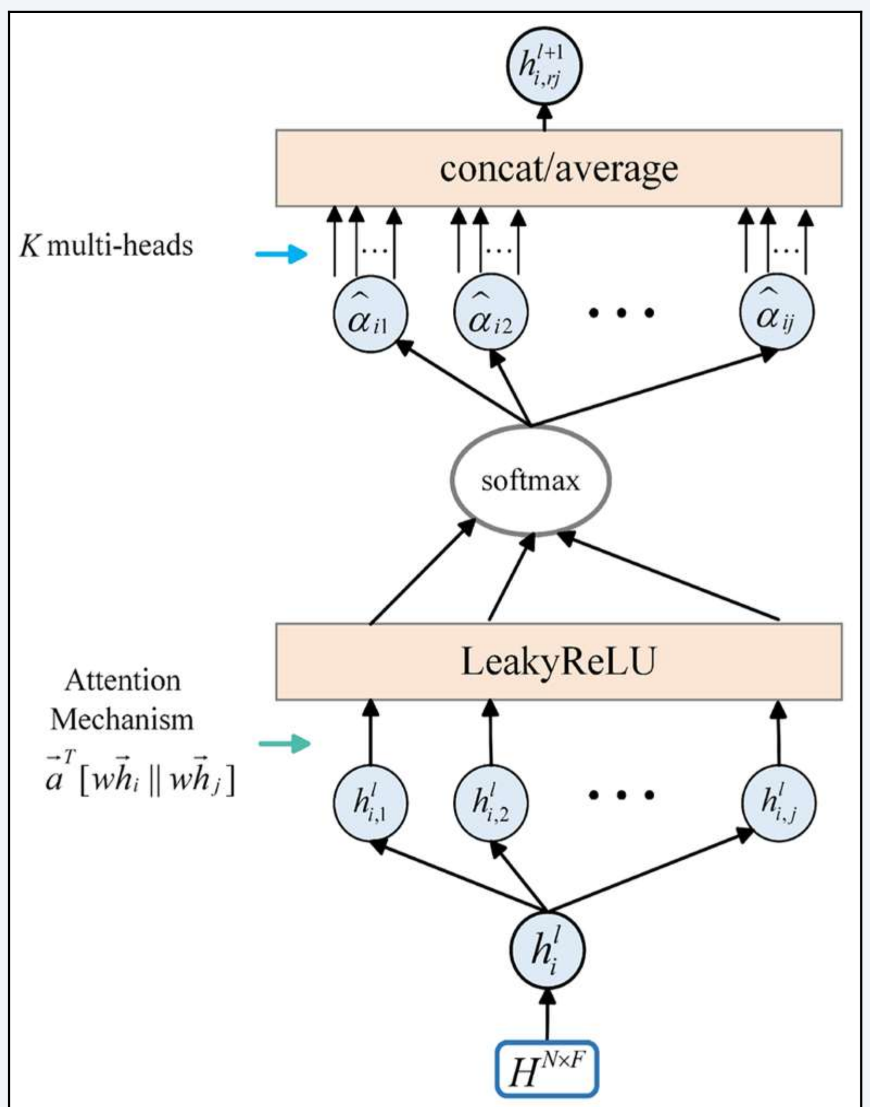

Fig. 3 Graph Attention Network framework Adaptive attention weights enable the model to focus on important neighboring nodes.

Combined with k-NN graph for local interaction modeling, Adaptive attention enables selective focus on important flow regions. This allows the network to better capture complex aerodynamic features by assigning different importance to neighboring nodes.

Results - Model Performance &Visualization

Method	Criteria category: ML-related (40%) – Accuracy(75%)					Physics(30%) – Physical Criteria				OOD generalization (30%) – OOD Accuracy(42%)					OOD generalization (30%) – OOD Physics(33%)				Global Score (%)
	$\overline{u}_x$	$\overline{u}_y$	$\overline{p}$	$\overline{\nu}_t$	$\overline{p}_s$	C_D	C_L	ρ_D	ρ_L	$\overline{u}_x$	$\overline{u}_y$	$\overline{p}$	$\overline{\nu}_t$	$\overline{p}_s$	C_D	C_L	ρ_D	ρ_L	
OpenFOAM	🟢	🟢	🟢	🟢	🟢	🟢	🟢	🟢	🟢	🟢	🟢	🟢	🟢	🟢	🟢	🟢	🟢	🟢	**82.5**
Baseline(FC)	🔴	🟠	🔴	🟢	🔴	🔴	🟠	🔴	🟠	🔴	🟠	🔴	🟢	🔴	🔴	🔴	🔴	🔴	**32.85**
Our model	🔴	🔴	🔴	🔴	🔴	🟠	🟠	🔴	🟠	🔴	🔴	🔴	🔴	🔴	🟠	🟠	🔴	🟠	**27.03**

- The model did not fully meet the accuracy requirements across all evaluation metrics.
- Large errors in turbulent viscosity and surface pressure caused inaccurate drag prediction, reducing the overall performance score.
- Lift prediction shows strong agreement with the ground truth, indicating that the overall aerodynamic behavior is captured.
- The model learns global flow patterns well but has difficulty predicting local physics-sensitive quantities.

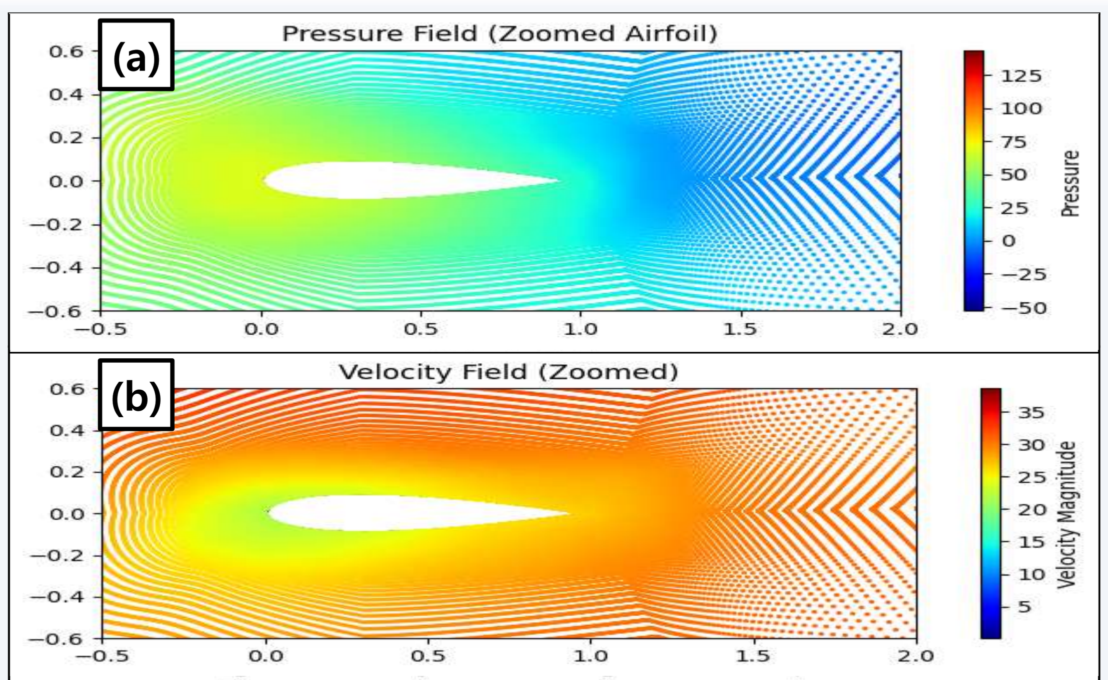

Fig. 4. Model predictions of pressure and velocity fields on the airfoil domain. (a) pressure field (b) velocity field

Conclusion

The GAT model captures the overall flow behavior but shows limitations in predicting physics-sensitive quantities such as drag. Errors in turbulent viscosity and surface pressure reduce the overall performance of the model. Future work will investigate alternative architectures and improved modeling strategies to achieve higher prediction accuracy.